

Subject Description Form

Subject Code	ME35001																								
Subject Title	Robotics and Computer Integrated Manufacturing																								
Credit Value	3																								
Level	3																								
Pre-requisite/ Co-requisite/ Exclusion	Prerequisite: Students should have basic knowledge of mechanical systems																								
Objectives	To provide students with the fundamental knowledge of advanced manufacturing process and to teach students how intelligence can be applied																								
Intended Learning Outcomes	Upon completion of the subject, students will be able to: - Understand the basic working principles of common manufacturing process - Explain the fundamental principles of Boolean logic, fuzzy logic (FL) systems, artificial neural networks (ANN). - Apply simple machine learning algorithms to intelligent control, planning and scheduling - Present effectively in completing written reports of laboratory work and the given task.																								
Subject Synopsis/ Indicative Syllabus	Manufacturing – Working Principle and Applications of Articulated, SCARA, Delta, Cartesian, and Polar robots, Traditional and modern manufacturing process Machine (Artificial) Intelligence – Boolean Logic; Fuzzy Logic; Regression; Classification; Supervised Learning; Unsupervised Learning, Deep Learning Applied AI for Manufacturing – Enhance Precision through Control, Planning and Scheduling for multi-agent systems																								
Teaching/Learning Methodology	Lectures aim at providing students with an integrated knowledge required for understanding and applying intelligence to enhance the efficiency and performance of robots for manufacturing (Outcomes a to d) Tutorials aim at enhancing students' analytical and problem-solving skills. Students will be able to solve real-world problems using the knowledge they acquired in the class. (Outcomes b to c) The project/experiment aims to have knowledge of computer simulations and hand-on experience on robots for manufacturing. (Outcomes a to d) <table border="1" style="margin-top: 10px;"> <thead> <tr> <th rowspan="2">Teaching/Learning Methodology</th><th colspan="4">Outcomes</th></tr> <tr> <th>a</th><th>b</th><th>c</th><th>d</th></tr> </thead> <tbody> <tr> <td>1. Lectures</td><td>√</td><td>√</td><td>√</td><td>√</td></tr> <tr> <td>2. Tutorials</td><td></td><td>√</td><td>√</td><td></td></tr> <tr> <td>3. Project or experiments</td><td>√</td><td>√</td><td>√</td><td>√</td></tr> </tbody> </table>	Teaching/Learning Methodology	Outcomes				a	b	c	d	1. Lectures	√	√	√	√	2. Tutorials		√	√		3. Project or experiments	√	√	√	√
Teaching/Learning Methodology	Outcomes																								
	a	b	c	d																					
1. Lectures	√	√	√	√																					
2. Tutorials		√	√																						
3. Project or experiments	√	√	√	√																					

Assessment Methods in Alignment with Intended Learning Outcomes	<table><tr><th rowspan="2">Specific assessment methods/tasks</th><th rowspan="2">% weighting</th><th colspan="4">Intended subject learning outcomes to be assessed</th></tr><tr><th>a</th><th>b</th><th>c</th><th>d</th></tr><tr><td>1. Assignments</td><td>10%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>2. Tests</td><td>20%</td><td>√</td><td>√</td><td>√</td><td></td></tr><tr><td>3. Projects</td><td>20%</td><td></td><td>√</td><td>√</td><td>√</td></tr><tr><td>4. Examinations</td><td>50%</td><td>√</td><td>√</td><td>√</td><td>√</td></tr><tr><td>Total</td><td>100 %</td><td colspan="4"></td></tr></table>	Specific assessment methods/tasks	% weighting	Intended subject learning outcomes to be assessed				a	b	c	d	1. Assignments	10%	√	√	√	√	2. Tests	20%	√	√	√		3. Projects	20%		√	√	√	4. Examinations	50%	√	√	√	√	Total	100 %				
	Specific assessment methods/tasks			% weighting	Intended subject learning outcomes to be assessed																																				
		a	b		c	d																																			
	1. Assignments	10%	√	√	√	√																																			
	2. Tests	20%	√	√	√																																				
	3. Projects	20%		√	√	√																																			
	4. Examinations	50%	√	√	√	√																																			
Total	100 %																																								
Explanation of the appropriateness of the assessment methods in assessing the intended learning outcomes:																																									
Overall Assessment: 0.50 × End of Subject Examination + 0.50 × Continuous Assessment																																									
The examination is adopted to assess students on their overall understanding and ability to apply the concepts. It is supplemented by continuous assessment including assignments, closed-book tests, and projects. The continuous assessment is aimed at enhancing the students’ comprehension and assimilation of various topics of the syllabus.																																									
Student Study Effort Expected	Class contact:																																								
	▪ Lecture				33Hrs.																																				
	▪ Tutorial				6Hrs.																																				
	Other student study effort:																																								
	▪ Reading and review				45Hrs.																																				
	▪ Coursework (assignments, project)				36Hrs.																																				
	Total student study effort				120Hrs.																																				
Reading List and References	1. Ray Asfahl, Robots and Manufacturing Automation, applications, Wiley, latest edition. 2. Wilson M, Implementation of Robot Systems, An Introduction to Robotics, Automation, and Successful Systems Integration in Manufacturing, latest edition M. W. Spong S. Hutchinson, and M. Vidyasagar, Robot Modeling and Control, Wiley, latest edition. 5. Campesato O., Artificial Intelligence, Machine Learning and Deep Learning, Mercury Learning & Information, latest edition																																								
Develop Date:	June 2024																																								